Functional Dependencies and Armstrong's Axioms

A Functional Dependency (FD) X -> Y states that if two tuples agree on attributes X, they must agree on attributes Y. X is called the Determinant.

Armstrong's Axioms are sound and complete inference rules for FDs:
1. Reflexivity: If Y is a subset of X, then X -> Y.
2. Augmentation: If X -> Y, then XZ -> YZ for any set of attributes Z.
3. Transitivity: If X -> Y and Y -> Z, then X -> Z.
Derived rules include Union (X -> Y and X -> Z implies X -> YZ) and Decomposition (X -> YZ implies X -> Y and X -> Z).

First (1NF), Second (2NF), and Third (3NF) Normal Forms

Normalization eliminates insertion, update, and deletion anomalies caused by data redundancy.

1NF: All attribute values must be atomic (no repeating groups, arrays, or nested tables).

2NF: Relation is in 1NF and every non-prime attribute is fully functionally dependent on every candidate key (no Partial Dependencies where a non-prime depends on a proper subset of a composite candidate key).

3NF: Relation is in 2NF and no non-prime attribute is transitively dependent on a candidate key (for every non-trivial FD X -> Y, either X is a super key or Y is a prime attribute).

Boyce-Codd Normal Form (BCNF) and Decomposition Criteria

A relation is in BCNF if for every non-trivial functional dependency X -> Y, X must be a Super Key. BCNF is strictly stronger than 3NF because it prohibits non-super key determinants even if Y is a prime attribute.

Decomposition Requirements:
1. Lossless-Join Decomposition: Decomposing R into R1 and R2 must satisfy that R1 intersect R2 -> R1 or R1 intersect R2 -> R2, guaranteeing no spurious tuples upon natural join.
2. Dependency Preservation: The union of functional dependencies in decomposed relations must logically imply all original dependencies in F+.